

Students will be assessed on their ability to:		Suggested experiments
59	define and use the concepts of emf and internal resistance and distinguish between emf and terminal potential difference	Measure the emf and internal resistance of a cell, e.g. a solar cell
60	investigate and recall that the resistance of metallic conductors increases with increasing temperature and that the resistance of negative temperature coefficient thermistors decreases with increasing temperature	Use of ohmmeter and temperature sensor
61	use $I = nqvA$ to explain the large range of resistivities of different materials	Demonstration of slow speed of ion movement during current flow
62	explain, qualitatively, how changes of resistance with temperature may be modelled in terms of lattice vibrations and the number of conduction electrons	

2.5 Nature of Light

This topic covers the wave/particle nature of light.

This topic may be studied either by using applications that relate to, for example, solar cells or by the historical study of the nature of light.

Students will be assessed on their ability to:		Suggested experiments
63	explain how the behaviour of light can be described in terms of waves and photons	
64	recall that the absorption of a photon can result in the emission of a photoelectron	Demonstration of discharge of a zinc plate by ultra violet light
65	understand and use the terms <i>threshold frequency</i> and <i>work function</i> and recognise and use the expression $hf = \phi + \frac{1}{2}mv_{max}^2$	
66	use the non-SI unit, the electronvolt (eV) to express small energies	
67	recognise and use the expression $E = hf$ to calculate the highest frequency of radiation that could be emitted in a transition across a known energy band gap or between known energy levels	
68	explain atomic line spectra in terms of transitions between discrete energy levels	Demonstration using gas-filled tubes
69	define and use radiation flux as power per unit area	
70	recognise and use the expression $\text{efficiency} = \frac{[\text{useful energy (or power) output}]}{[\text{total energy (or power) input}]}$	
71	explain how wave and photon models have contributed to the understanding of the nature of light	
72	explore how science is used by society to make decisions, for example, the viability of solar cells as a replacement for other energy sources, the uses of remote sensing	